

# Dive Buddy Alert Band

Paired underwater calls with visible distance feedback.



## Feel the call. See the distance.

A slip-on wristband concept combines a paired tactile call with a small display. Vibration draws attention. The screen shows call status and a fresh distance estimate during ranging.

### Rev C display concept

A glanceable display sits beside the physical top button. Both divers can see the current separation and communication state. Inductive charging through the sealed cap and the removable module remain part of the design.

The render above shows the original CAD baseline. Rev C adds a display appearance and interaction study. Screen dimensions, enclosure integration, power consumption and underwater legibility require engineering definition.

**One wrist transducer provides distance, not direction. The band is an aid to buddy procedures, never a safety guarantee.**

# User interaction

Proposed surface pairing and underwater button sequence.

## 1 / Pair on the surface

Hold both buttons for 3 seconds. Each screen shows the paired buddy and local battery status. Address allocation and nearby-pair isolation require firmware definition.

## 2 / Call

A short press on a ready band calls the paired buddy. B receives three alert pulses and a visible incoming call. A sees Calling, followed by Received by device only after a packet reply.

## 3 / Acknowledge

B presses once to acknowledge the incoming call. A then sees Buddy acknowledged. A device reply confirms message delivery only. The separate button press communicates the buddy's response.

## 4 / Range

A holds for 1 second to start ranging. Both screens show the latest valid distance and a trend derived from successive readings. A can use the proposed distance cadence alongside the screen.

## 5 / Stop

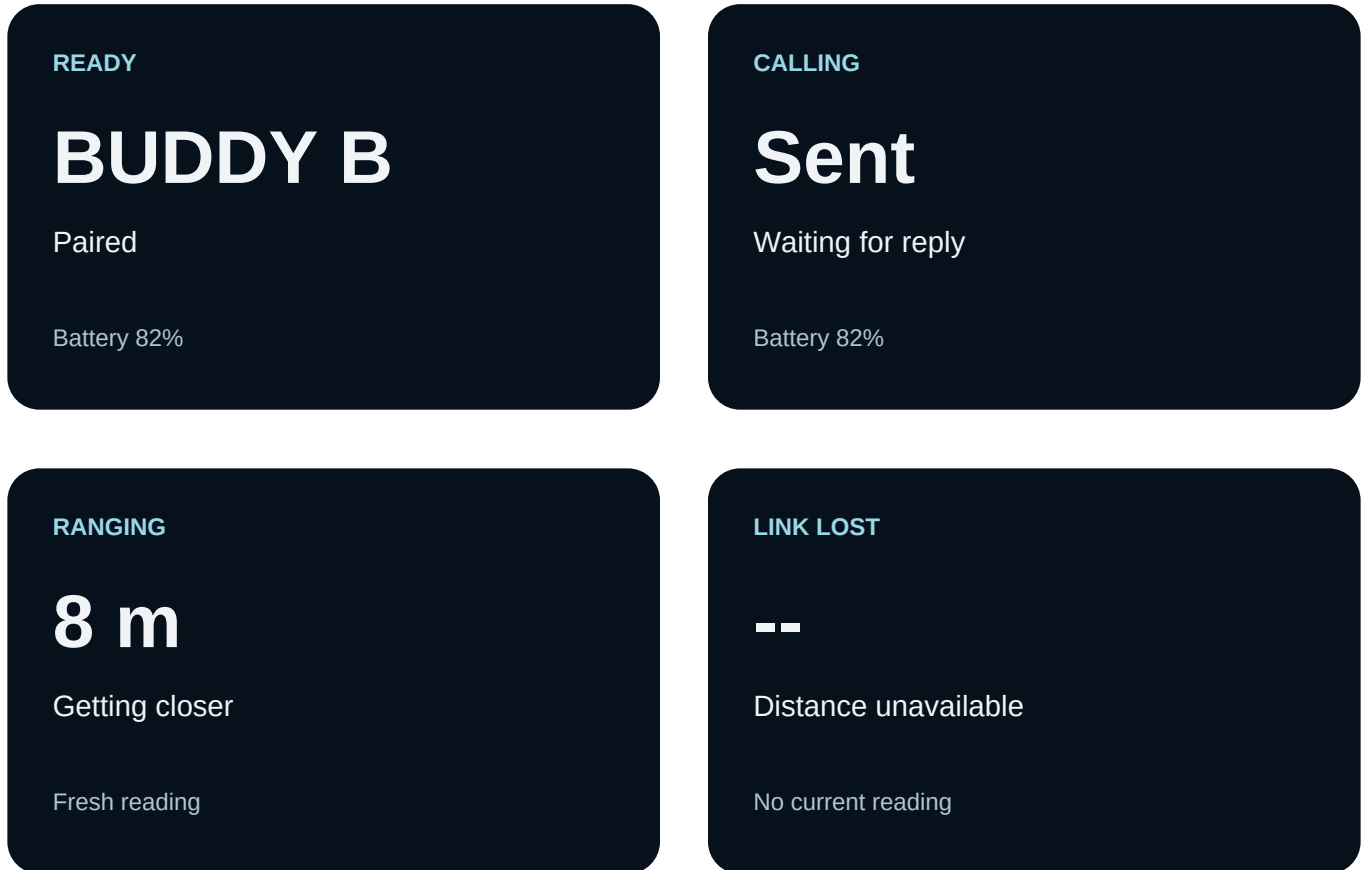
During ranging, a short press on either band requests Stop. The proposed five-minute timeout and one-second burst then stop below 2 m remain. Neither a near reading nor an automatic stop confirms reunion.

## 6 / Charge

Place the bands on an inductive dock. The display shows local charging status. There is no charging connector on the band. Dock alignment, thermal limits and charge completion require definition.

# The display

Proposed information hierarchy. Illustrative values and enlarged screen studies.



## A distance requires a fresh measurement

During ranging, the distance takes visual priority. A trend appears only after enough valid readings. A missing or stale reply clears the distance and trend and stops proximity cadence. Reacquisition begins with a fresh reading, without reusing an old trend.

Calling, device receipt and buddy acknowledgement remain separate states. Pair identity and battery stay available. Text accompanies status colors. The display uses the physical button rather than an underwater touchscreen.

These screens specify proposed content and behavior. The 8 m distance and 82% battery are synthetic. Pixel layout, refresh timing, brightness, window size and underwater readability are not established.

# Distance feedback

Acoustic range, visual trend and optional caller cadence.

**15-20 m / valid reply** - One pulse every 4 seconds.

**10-15 m** - One pulse every 2 seconds.

**5-10 m** - One pulse every second.

**2-5 m** - A double pulse every 0.5 seconds.

**Below 2 m** - One-second burst, then stop. Proximity does not confirm visual contact.

**Lost or stale reply** - Distance unavailable. Clear the distance and trend and stop proximity pulses.

## Distance and orientation

The proposed single transducer estimates separation. A range of 8 m does not establish whether the buddy is left, right, above or below. Relative bearing requires a separate sensing architecture. Exchanging device headings alone would not establish the bearing between divers, and wrist heading need not match swimming direction.

## The acoustic exchange

Approximately 65 kHz ultrasonic two-way time-of-flight. Proposed calculation:  **$d = (\text{round-trip time} - \text{fixed reply delay}) \times \text{sound speed} / 2$** . At 1,500 m/s with an assumed 2 ms reply delay, 20 m gives approximately 26.7 ms propagation round trip and 28.7 ms total exchange time.

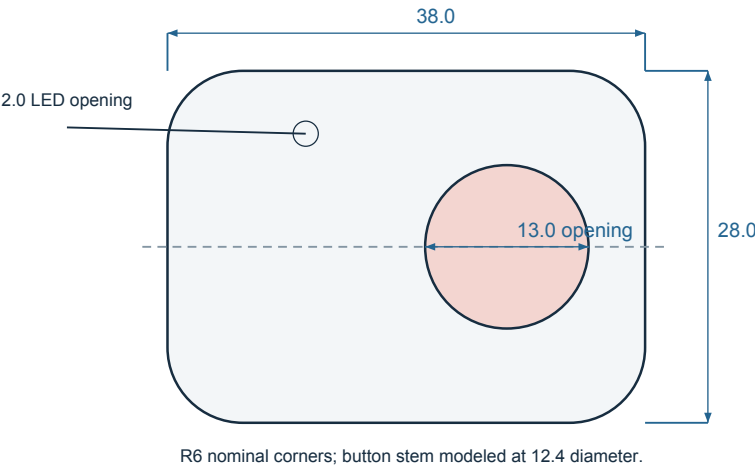
20 m range, approximately  $\pm 0.3$  m distance performance and nominal 1 Hz exchange are targets or assumptions. No measured performance is supplied. Timing resolution does not establish accuracy. Multipath, body shadowing and environmental variation remain unresolved.

# Original CAD dimensions

Rev A baseline geometry. Display integration has no released dimensions.

## MODULE / TOP VIEW

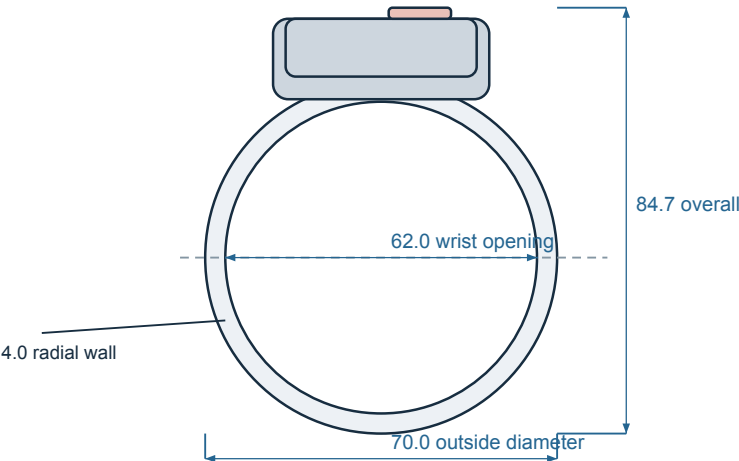
Housing envelope excludes button protrusion.



DBA / REV B | NOMINAL mm | CONCEPT ONLY - NOT TO SCALE

## BAND / FRONT VIEW

Existing source geometry, with the module assembled.  
24.0 band width along wrist axis; 33.0 cradle width.



DBA / REV B | NOMINAL mm | CONCEPT ONLY - NOT TO SCALE

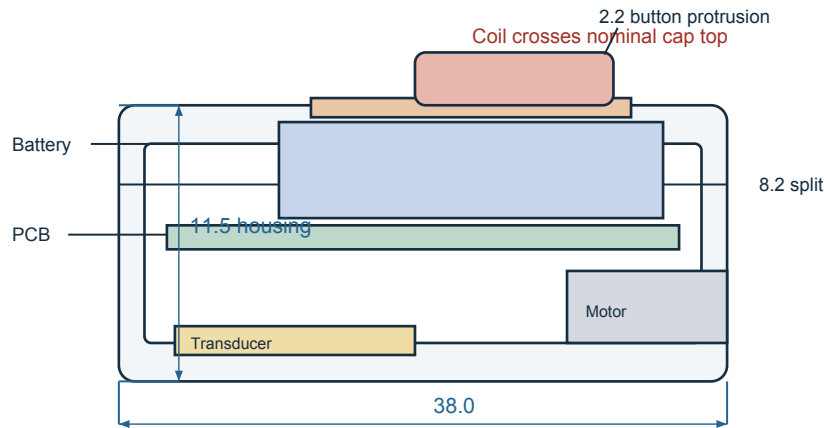
The nominal 38 x 28 x 11.5 mm housing excludes the 2.2 mm modeled button protrusion. The source gives 84.7 mm overall height including the button; the original DBA-001 sheet shows 88.5 mm. These values do not establish the envelope of the Rev C display variant.

# Display integration & internal layout

Existing component layout and additional interfaces for Rev C.

## MODULE / SCHEMATIC SECTION

Internal packaging is unresolved. This view does not certify fit.



Cap/battery overlap, coil clearance and switch/button stack require reconciliation.

DBA / REV B | NOMINAL mm | CONCEPT ONLY - NOT TO SCALE

## A sealed display window

The proposed display adds a window, optical stack, electrical connection and power load beside the top button. Window sealing, pressure deflection, impact resistance, condensation and readability need definition. The coil, display electronics and window must coexist within a revised packaging design.

## Original packaging conflicts

The source coil occupies 11.0-11.8 mm above the housing base, while the housing ends at 11.5 mm. The battery occupies 6.8-10.8 mm and extends into the nominal cap region. Button/switch clearances, wire routing, ferrite, adhesives and battery allowance remain unresolved.

The module remains removable from the silicone band. The sealed enclosure is not specified as user-openable. Inductive charging avoids an exposed band connector, with alignment and thermal design as trade-offs. No CAD geometry or original mesh has changed in Rev C.

# Engineering specification

Register 1-5 of 24. Design decisions, performance targets and pending definition.

## Product scope / Fixed decision

One-to-one buddy attention and distance-feedback concept; an aid to buddy procedures, never a safety guarantee.

**Pending definition:** No promise of reunion, emergency rescue, bearing or navigation.

**Discipline:** Systems / product

## Form and service / Fixed decision

Slip-on silicone band; sealed module removable from the band; button on top.

**Pending definition:** Retention force, release method, snag loads and actual wrist/cuff fit remain open. Removal from the band does not mean the sealed enclosure is user-openable.

**Discipline:** Mechanical / industrial design

## Charging interface / Fixed decision

Inductive charging through the sealed cap; no charging port on the band.

**Pending definition:** Dock alignment, spacing, thermal limits, wet/salt-contaminated charging behavior and charge indication are unspecified. The dock may have its own power connector.

**Discipline:** Electronics / mechanical

## Acoustic method / Fixed decision

Approximately 65 kHz ultrasonic two-way time-of-flight; one wrist transducer.

**Pending definition:** Transducer construction/resonance, directivity, drive level, link budget and receiver sensitivity need engineering definition. Component names in Rev A are candidates, not selections.

**Discipline:** Underwater acoustics

## Distance and timing / Proposed target

20 m range; approximately  $\pm 0.3$  m distance target; nominal 1 Hz exchange with assumed fixed 2 ms reply delay.

**Pending definition:** No measured performance supplied. Accuracy, resolution, latency and availability need separate definitions. Use  $d = (RTT - \text{reply delay}) \times \text{sound speed} / 2$ ; timing precision alone does not establish accuracy.

**Discipline:** Acoustics / firmware

# Engineering specification

Register 6-10 of 24. Design decisions, performance targets and pending definition.

## Operating envelope / Proposed target

Rev A proposes 50 m operating depth, four 60-minute dives per charge and about 28 g mass. These are baseline targets; display-variant mass and runtime are unestimated.

**Pending definition:** No depth rating or certification established. Define immersion duration, temperature, salinity, pressure convention, storage conditions and intended environments. The Rev A 6 bar note is ambiguous without gauge/absolute pressure.

**Discipline:** Systems / mechanical

## Pairing / Rev C proposal

Hold both buttons 3 s on the surface; show the paired buddy and Ready on both displays. The 16-bit pair code remains a Rev A protocol assumption.

**Pending definition:** Address allocation, cross-pair isolation and re-pairing protocol remain open. The simulation stages mutual pairing; it does not discover real devices.

**Discipline:** Firmware

## Call and acknowledgement / Rev C proposal

Short press calls the paired buddy. B receives three alert pulses and an incoming-call screen; B short presses to acknowledge. A distinguishes Calling, Device received and Buddy acknowledged.

**Pending definition:** Transport acknowledgement confirms the device received a packet, not human attention. Retry limits, collision handling and delayed or duplicated messages need protocol definition.

**Discipline:** Firmware / interaction design

## Ranging and stop / Rev C proposal

Caller holds 1 s to request ranging after the call. Both displays show fresh distance when available; caller cadence is optional. During ranging, a short press on either band stops; nominal 5-minute timeout.

**Pending definition:** Remote stop requires confirmation. The demonstration separates local stop from unconfirmed delivery when the link is unavailable. Both displays and synchronization remain proposed firmware behavior.

**Discipline:** Firmware / interaction design

## Haptic vocabulary / Rev C proposal

Attention uses three pulses. Optional caller distance cadence retains Rev A: 15-20 m pulse/4 s; 10-15 m pulse/2 s; 5-10 m pulse/s; 2-5 m double pulse/0.5 s; below 2 m a 1 s burst then stop.

**Pending definition:** Display provides distance without requiring cadence interpretation. Actual intensity, perceptibility through cuffs, hysteresis and waveforms remain unresolved. Near does not mean visual contact or reunion.

**Discipline:** Interaction design / firmware



# Engineering specification

Register 11-15 of 24. Design decisions, performance targets and pending definition.

## Lost link / Rev C proposal

Missing or stale replies clear the current distance and trend and suppress proximity pulses. Display reads Distance unavailable; a fresh valid update is required to restore a number.

**Pending definition:** Demonstration freshness timeout and quality switches are illustrative. Production age limits, validation, outlier handling and reacquisition require acoustic and firmware definition.

**Discipline:** Systems / firmware

## Orientation and echoes / Open question

Body shadowing, wrist orientation, obstructions, other pairs and multipath may affect reception or range.

**Pending definition:** Define accepted environments, range-quality reporting, outlier handling and stale-reading policy. No directional arrow is provided to the diver.

**Discipline:** Underwater acoustics

## Power budget / Rev A baseline

Original 150 mAh, 3.7 V cell envelope; approximately 0.4 mA listening, 12 mA ranging and 2 h charging assumptions exclude the new display.

**Pending definition:** Display, illumination and control loads need a revised budget. Rev A stated currents imply about 5.5 mAh for its stated duty cycle before omitted loads, not about 100 mAh. Neither establishes runtime.

**Discipline:** Electronics

## Battery and charge behavior / Rev C proposal

Visible local battery status and a low-battery indication; illustrative demo values do not represent measured charge or a validated reserve.

**Pending definition:** Low-battery thresholds, reserve behavior, cell protection, cold-water effects, aging and charging completion remain to be defined. Battery percentages alone do not establish remaining dive time.

**Discipline:** Electronics / firmware

## Housing and seal / Rev A assumption

Rev A baseline housing: 38 x 28 x 11.5 mm, 1.6 mm nominal wall, 0.6 mm acoustic membrane. The display appearance study does not establish a screen/window fit.

**Pending definition:** Material choice, joint, sealing details, pressure deflection, corrosion compatibility, leak paths, light-pipe/button sealing and acoustic coupling remain unresolved. No manufacturing tolerances are released.

**Discipline:** Mechanical / acoustics

# Engineering specification

Register 16-20 of 24. Design decisions, performance targets and pending definition.

## Internal packaging / Open question

Original source places the coil above the cap envelope and the battery into the cap region. Rev C adds a display appearance/UI study without changing this baseline CAD.

**Pending definition:** Display window, electronics, illumination, coil/ferrite spacing, pressure deflection, thermal behavior and button access need new integration. No display fit, seal or structural performance is established.

**Discipline:** Mechanical / electronics

## Band and button / Rev A assumption

62 mm relaxed wrist opening; 24 mm band width; 4 mm radial wall; 13 mm nominal button aperture; 3-4 N actuation target.

**Pending definition:** 95 mm stretched opening, one-size fit and glove operation are assumptions.  $BTN\_H = 2.2$  mm is modeled protrusion, not demonstrated switch travel. The visible stem is 12.4 mm in the source; flange is 16 mm.

**Discipline:** Industrial design / mechanical

## Dimensions and drawing authority / Open question

Source gives 84.7 mm nominal assembled height including button; original DBA-001 shows 88.5 mm. Use the Rev B annotated concept views for presentation.

**Pending definition:** Rev A's blanket  $\pm 0.2$  tolerance and drawing scale are not a release specification.  $SPLIT\_Z = 3.2$  is used with +5.0 in the source, yielding an 8.2 mm base. Do not infer parameter behavior from names alone.

**Discipline:** Mechanical

## Commercial case / Hypothesis

Proposed first audience: recreational buddy pairs, reached through instructors and dive shops; paired sale with shared dock.

**Pending definition:** No customer interviews, purchase commitments or defensible novelty evidence supplied. The value of an additional wrist device and the display/haptic combination remains a hypothesis.

**Discipline:** Product / commercial

## Economics / Rev A desk estimate

Original display-free concept: approximately \$43/band, \$86/pair at 5,000 pairs,  $\pm 30\%$ ; \$179-199/pair retail target. Rev C display cost is unestimated.

**Pending definition:** Baseline only, not a quote or current display-variant cost. Display and revised enclosure/power integration need a new estimate. Tooling, NRE, freight and duty are excluded; margins remain open.

**Discipline:** Commercial / manufacturing

# Engineering specification

Register 21-24 of 24. Design decisions, performance targets and pending definition.

## Project status / Concept revision

Concept by Rafael. Rev C proposes a hybrid display/haptic interface; original parametric layout remains available as a baseline. No engineered display assembly is supplied.

**Pending definition:** The screen mockups and material treatment illustrate intended information and placement. Feasibility, packaging, readability, customer value and manufacturing readiness remain unestablished.

**Discipline:** Project lead

## Display and readability / Rev C proposal

Sparse screens for pairing, readiness, call delivery, human acknowledgement, fresh estimated distance, unavailable data and battery. Whole-metre demo values avoid suggesting validated decimal precision.

**Pending definition:** Display technology, usable area, viewing angle, contrast, illumination and legibility underwater are undefined. Enlarged screen mockups are not physical size or readability proof.

**Discipline:** Interaction design / electronics / mechanical

## Direction and orientation / Outside current capability

One-transducer two-way timing provides range. No buddy bearing, relative heading or depth difference is measured or displayed.

**Pending definition:** A directional buddy-finding feature requires separately defined sensing and estimation. Exchanging device headings alone does not determine relative location; wrist heading is not necessarily swimming direction.

**Discipline:** Systems / underwater acoustics

## Distance trend / Rev C proposal

Nearer / farther is an optional textual comparison of successive fresh distance estimates. It describes separation change, not a route or direction to swim.

**Pending definition:** The simulator uses controlled values. Sampling interval, deadband, stability, quality gating and outlier rejection need definition before a measured trend can be shown.

**Discipline:** Firmware / interaction design

# Paired kit & commercial assumptions

Original cost basis and the scope of the display revision.

## Proposed offer

Two bands with one shared inductive dock for recreational buddy pairs. Instructors and dive shops are potential channels. Customer demand and the value of a separate wrist device remain unvalidated.

## Rev A baseline economics

Rev A gives a retail target of **\$179-199 per pair** and a desk cost estimate of approximately **\$43 per band / \$86 per pair** at 5,000 pairs, with **±30% uncertainty**. The estimate allocates the dock across the pair. It excludes tooling, non-recurring engineering, freight and duty.

**The display variant has no revised cost estimate or retail target. The screen, window, seals, control electronics and power requirements change the cost basis. Rev A figures do not price the Rev C design.**

## Performance scope

The original four 60-minute dives per charge, approximately 28 g assembled mass and nominal housing envelope do not establish the display variant's runtime, mass or size. The display load must enter a complete energy budget. The original energy arithmetic and packaging conflicts remain recorded in the specification.

## Development status

Rev C defines a proposed interface and appearance study. The original eleven-part CAD remains the mechanical baseline. Acoustic performance, manufacturing readiness, customer demand, certification and intellectual-property clearance remain unestablished. No hardware program is commissioned by this presentation.

# Design files

Current presentation and original parametric reference.

## **DBA\_Interactive.html**

Rev C product presentation with a display appearance study, interactive two-diver demonstration, original 3D CAD, dimensions and the full specification. Screen values are simulated. Three.js loads from its existing CDN.

## **DBA\_Design\_Package\_RevC.pdf**

Current concept, proposed display and button behavior, technical specification and engineering review notes.

## **DBA\_Product\_Overview\_RevC.pptx**

Editable product overview with speaker notes. Illustrative screens communicate the proposed interface.

## **DBA\_Annotated\_Views\_RevB.pdf**

Three unchanged annotated views of the original CAD. Nominal dimensions are baseline references, without manufacturing tolerances or display integration.

## **cad/, drawings/ and renders/**

Original STEP, STL, DXF, drawings and render assets. The original dimension discrepancy and unresolved layout remain recorded in this brief.

## **dba\_cad.py, render.py, drawing\_sheet.py**

Original parametric and rendering source. The original output path points to /home/claude/out. Future authorized regeneration requires the correct local output path.

## **presentation/**

Editable presentation source and requirements register. The original int16 mesh payload remains intact.

## **Rev A and Rev B references**

Earlier documents remain historical references. Rev C governs the proposed display interaction. Original drawings govern only the unchanged CAD baseline.

**Future geometry changes begin in dba\_cad.py, followed by regeneration of CAD, renders, the drawing sheet and embedded meshes. Rev C changes the interface concept and presentation only.**